

Lie detection protocol

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1 Introduction

In this document, we identify the EEG channels and experimental methods used to detect deception in subjects. Many previous studies have utilized EEG channels without offering detailed explanations. For this project, our goal is to understand the neural mechanisms underlying deception in social contexts and develop our own process for creating a lie detector database. Subsequently, we will employ data processing methods to identify the features that will ultimately enable us to classify the data effectively.

2 EEG Channels

Few documents have been published on this subject, and most of them do not explain why specific EEG channels are used ([2], [4], [3], [1], [5], [8], [10], [7]).

2.1 Visualization of deception with fMRI

In various studies, researchers have gained a better understanding of the brain mechanisms involved in deception in strategic interactions. The initial study conducted in 2015 [9] identified key brain regions associated with deception. Other research has highlighted additional brain areas involved in deceptive behavior. These key brain regions include the right temporoparietal junction, the left temporoparietal junction, the retrosplenial cortex, the left precuneus, and the right superior temporal sulcus. This information was extracted using functional magnetic resonance imaging (fMRI). See Figure 1 for activated brain areas during deception.

2.2 Selecting the EEG Channels

We will use all EEG channels in our experiment. Referring to the EEG model shown in Fig: 10-20 EEG model for experimentation, we identify the appropriate sensors for our experiment.

The EEG signal is recorded to capture information from specific brain regions, including the following:

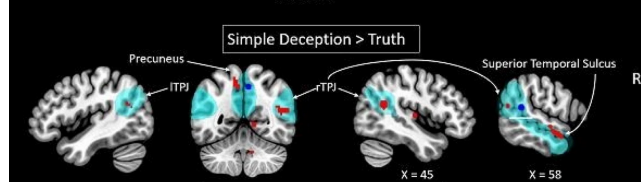


Figure 1: Deception visualization with fMRI

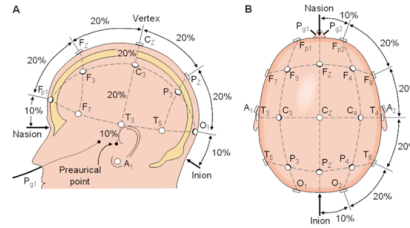


Figure 2: EEG 10-20 Model for experimentation

- Right TPJ
- Left TPJ
- Retrosplenial cortex
- Left precuneus
- Right STS

Thanks to the identification of these area on MRI, we will be able to use them during the "DataProcessing" stage to focus our research on specific zones or channels.

3 Experimentation

3.1 Experimentation for the fMRI

In the study [11] [9] a particular experimentation has been used. This experimentation is the Sender Receiver game.

The sender-receiver game is a complex game designed to detect truth and two types of lies: simple deception and sophisticated deception(see Figure 3). Four possible results are obtained: sophisticated deception, truth, simple deception, and not classifiable. In our study, we focus on simple deception. That's why we develop a simplified lie detection game with two possible result like we can see Figure 4.

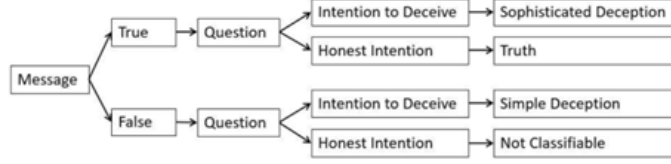


Figure 3: Sender-Receiver Game

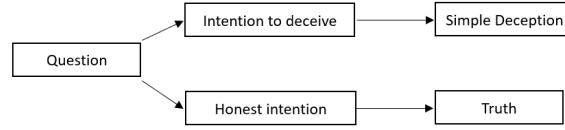


Figure 4: Lie Detection Game

3.2 Description of the Experiment

Participants are presented with two distinct sets of images alternately on the screen. Each image is displayed for 7 seconds. Within this timeframe, participants must decide whether to tell the truth or deceive by selecting one of two corresponding buttons. These buttons may or may not align with the content of the displayed image, we can see Figure 5 the Experimentation graph. The interface for this game is shown in Figure 6.

To help with decision-making and reduce interference from other brain activities, participants are guided to use specific keys on the keyboard to make their selections. This method is intended to decrease unnecessary cognitive processes and simplify the neural activity related to decision-making, recognizing that this activity primarily occurs in the cortex.

Afterward, the computer compares these brain signals with what it knows to be the correct answers. This helps us spot any differences between what the participants' brains are doing and what they should be doing, making our lie detection more accurate and allowing us to build a database of brain activity.

3.3 Result

At the end of my internship, we have recorded the data of 3 different subjects. for a total of 568 epochs (279 lie and 289 true).

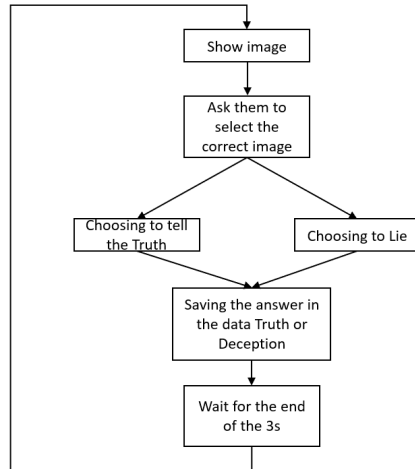


Figure 5: Experimentation graph



Figure 6: Lie Detector Game

4 Data Processing

4.1 Preprocessor

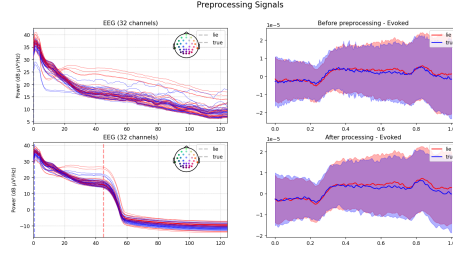


Figure 7: Preprocessing raw data

Our Preprocessing chain is built as follows:

- Bandpass filter from 0.5 Hz to 45Hz
- Remove bads channels
- Artifact Subspace Reconstruction (ASR)

The figure below 7 shows our data before and after this processing chain. The left-hand side of the figure shows the frequency representation of the signal and the right-hand side the time representation.

4.2 ERN

4.2.1 Analysis

We use an ERN representation of our EEG data to obtain a clear representation of the related epochs and to see if we can find any similarities. We will present the expected results found in the thesis below and then compare them with our own findings.

As discussed in [6], the following image of ERN was found (8). We have attempted to reproduce this with our data to observe whether the same phenomenon can be detected.

On the following images (9,10,11,12): red represents the related epochs, and blue represents the average true epochs across the channels.

4.2.2 Conclusion

We can identify similarities between the research findings and the extraction of our data from our experiments. This suggests that the research avenue is promising, and it would be beneficial to identify the corresponding features. Ultimately, this will enable us to perform classification effectively.

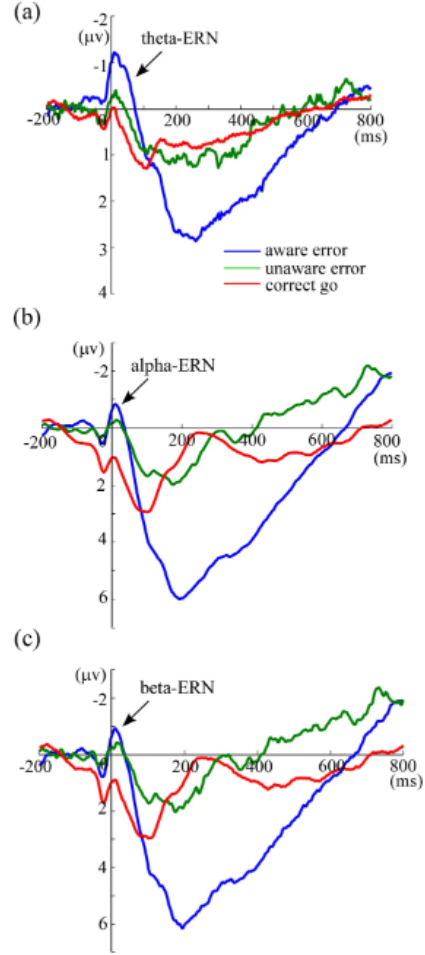


Figure 8: ERN representation example

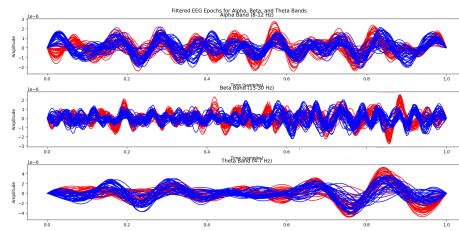


Figure 9: ERN representation subject 1 run 1

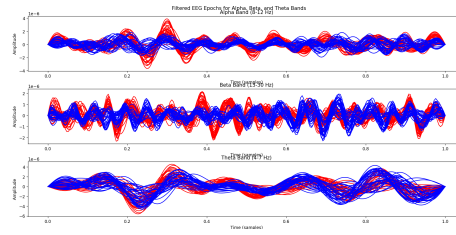


Figure 10: ERN representation subject 1 run 2

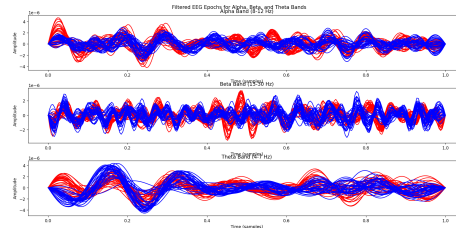


Figure 11: ERN representation subject 1 run 3

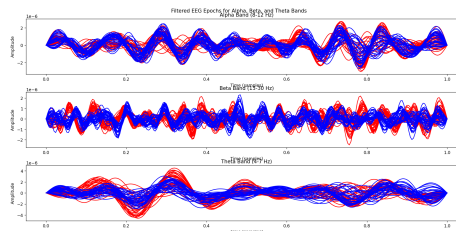


Figure 12: ERN representation subject 1 run 4

5 Others analysis

We tried a few other analyses that didn't give good results. These included looking at the timing of specific channels and using source localization along with feature selection. We will go over these methods in more detail in the next sections.

5.1 Temporal representation of specific channel

Based on previous studies, we are now able to concentrate on specific channels. This targeted approach will allow us to gain a clearer understanding of the dynamics and activities occurring in these particular areas. Following this, we will display the PSD representations associated with both the 'lie' and 'true' conditions for the CP5, CP6, and CZ channels.

The lack of favorable results may be attributed to the broad time window from -2s to 1s, as well as the averaging across channels, which might obscure finer details. However, we do observe notable cerebral activity starting after 0s, corresponding to the marker emitted when the subject presses the button. This indicates that there is indeed specific neural activity associated with the task

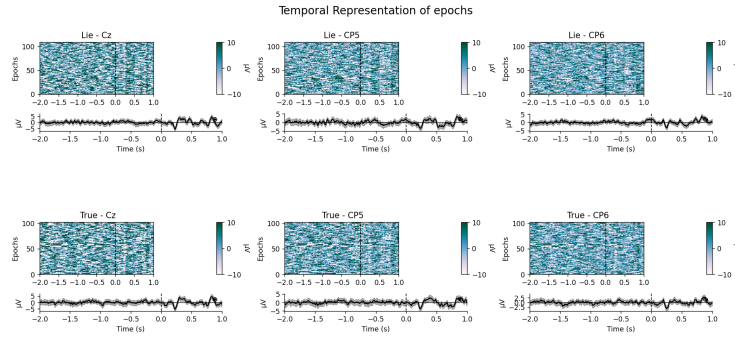


Figure 13: Power spectral density of 3 channels for lie and true epochs

5.2 Source Localisation and features selection

The analysis of ERN (Error-Related Negativity) seems to be the most promising approach, especially considering that we have already explored advanced techniques such as source localization. This method involves extracting over 8,000 sources. Features were derived from this extensive dataset. We then ranked these features by their importance, as illustrated in the following figure 15 14. Despite this, the approach ultimately proved to be non-viable

The maximum importance value for the frequency-based features is 0.028, which is not considered significant. Similarly, the maximum importance value for the time-based features is 0.0200.

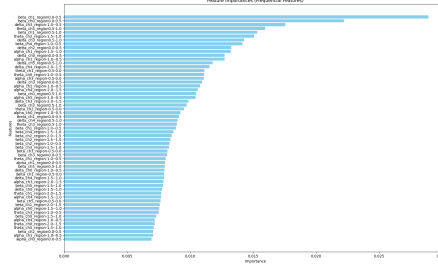


Figure 14: Importances of the frequencial features

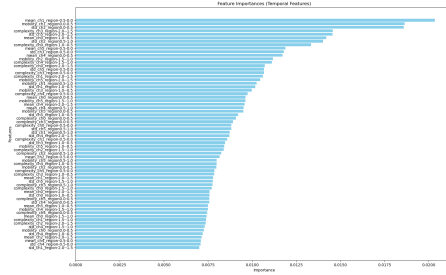


Figure 15: Importances of the temporal features

Even when considering the top features, we are unable to identify any particularly activated regions using this method.

6 Conclusion

To conclude, Based on our findings, it is evident that distinguishing between lies and truths, or more precisely, between correct and incorrect responses, is a challenging task. The methods we explored, including large-scale data analysis such as source localization and the extraction of both temporal and frequency-based features, turned out to be too unpredictable and complex for effective classification.

If I had more time to continue with this project, I would shift my focus to analyzing Error-Related Negativity (ERN) signals with a more refined approach. Specifically, I would concentrate on a smaller number of epochs and use only a few targeted channels. The CP5, CP6, and CP7 channels, in particular, are the most relevant because they are closest to the brain regions identified as significant in our EEG data.

By narrowing down the analysis to these specific channels and epochs, we could potentially improve the clarity and accuracy of our results. This more focused approach might provide better insights into the processes underlying the distinction between correct and incorrect responses, ultimately leading to

more reliable conclusions.

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